

# Notes de Théorie Algébrique des Nombres

GitHub Repository: [Oxke/appunti/AlgebraSuperiore](https://github.com/Oxke/appunti/AlgebraSuperiore)

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# Capitolo 1

## Number fields and their rings of integers

### 1.1 Integrality

#### Definizione 1

Take  $A \subseteq B$  an inclusion of domains. An element  $x \in B$  is said to be **integral over**  $A$  if it is a root of a monic polynomial in  $A[X]$ .

The ring  $B$  is said to be **integral over**  $A$  if all elements of  $B$  are integral.

**Esempio 2.** The complex numbers  $\overline{\mathbb{Z}}$  that are integral over  $\mathbb{Z}$  are called the *algebraic integers*.

**Esempio 3.** Let  $d \neq 0, 1$  squarefree. Then

$$\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Z}\}$$

is integral over  $\mathbb{Z}$ . Indeed if  $x = a + b\sqrt{d}$ , then  $(x - a)^2 - b^2d = 0$

**Esempio 4.** If  $\zeta = e^{\frac{2\pi i}{n}}$  then  $\zeta^n = 1$ , thus  $\zeta \in \overline{\mathbb{Z}}$

**Proposizione 5.** Let  $A$  be a UFD. Let  $k = \mathcal{Q}(A)$  the fraction field of  $A$ . An element  $x \in K$  is integral over  $A$  if and only if  $x \in A$

*Dimostrazione.* Write  $x = \frac{p}{q}$ , with  $p, q \in A$  and  $\text{mcd}(p, q) = 1$ . Then in  $x$  is integral over  $A$ , it satisfies an equation

$$x^d + a_{d-1}x^{d-1} + \cdots + a_0 = 0 \quad a_i \in A \quad \forall i$$

Then

$$\begin{aligned} \left(\frac{p}{q}\right)^d + a_{d-1}\left(\frac{p}{q}\right)^{d-1} + \cdots + a_0 &= 0 \\ \implies p^d + a_{d-1}p^{d-1}q + \cdots + a_0q^d &= 0 \end{aligned}$$

hence  $q|p^d$  but they are coprime, hence  $q \in A^*$  and thus  $x \in A$  □

Conversely, if  $x \in A$ , it is a root of  $X - x$ .

**Corollario 6.**  $\mathbb{Q} \cap \overline{\mathbb{Z}} = \mathbb{Z}$

**Proposizione 7.** Let  $A \subseteq B$  inclusion of domains. Fix  $b_1, \dots, b_n \in B$  not all zero. Set  $P = \sum_{i=1}^n Ab_i$  and take  $b$  such that  $bP \subseteq P$ .

We can write

$$bb_i = \sum_{j=1}^n a_{ij}b_j \quad a_{ij} \in A$$

Then introduce  $M = (a_{ij}) \in \mathcal{M}_n(A)$  and let

$$\chi_M(T) = \det(TI_n - M)$$

then  $\chi_M(b) = 0$

*Osservazione.*  $M^t$

*Dimostrazione.* Set  $N = bI_n - M$ . We have

$$N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

So

$$\det N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = ({}^t\text{Conj} N) N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

Hence, since there exists  $i_0$  with  $b_{i_0} \neq 0$ ,

$$\det N = 0 = \chi_M(b)$$

□

**Esempio 8.** Let  $A = \mathbb{Z}$ ,  $B = \mathbb{C}$  If we let  $b_1 = 1, b_2 = \sqrt[3]{2}, b_3 = \sqrt[3]{4}$ . Then we can take  $b = 1 + 3\sqrt[3]{2} - \sqrt[3]{4}$

**Corollario 9.** Let  $A \subseteq B$  inclusion of domains. Take  $x_1, \dots, x_r \in B$ . Then  $x_1, \dots, x_r$  are integral over  $A$  if and only if  $A[x_1, \dots, x_r]$  is finitely generated as an  $A$ -module.

*Dimostrazione.*

$\Rightarrow$  Assume  $x_1, \dots, x_r$  are integral over  $A$ . By induction, we may and do assume that  $r = 1$ . Take  $\mu \in A[X]$  monic with  $\mu(x_1) = 0$ . Take  $f(x_1) = x \in A[x_1]$ , with  $f \in A[X]$ . Since  $\mu$  is monic, we can write

$$f = \mu Q + R$$

with  $\deg R < \deg \mu$  and  $Q, R \in A[X]$ . Finally,  $f(x_1) = R(x_1) \in \sum_{i=1}^{\deg \mu - 1} Ax_1^i$  hence  $A[x_1]$  is finitely generated.

$\Leftarrow$  Assume  $A[x_1, \dots, x_r]$  is finitely generated as an  $A$ -module. Let  $b_1, \dots, b_n$  be generators and set  $b = x_i$ . Apply the previous proposition to this setting and get a monic  $\chi \in A[X]$  with  $\chi(x_i) = 0$ . Hence,  $x_i$  is integral over  $A$ .

□

**Corollario 10.** Let  $A \subseteq B$  inclusion of domains. Let  $\overline{A}$  be the set of elements of  $B$  that are integral over  $A$  then  $\overline{A}$  is a subring of  $B$

*Dimostrazione.* Take  $x, y \in \overline{A}$  and consider  $C = A[x, y]$ . Then  $C$  is finitely generated as an  $A$ -module. But  $C = A[x, y, x+y, xy]$ , hence  $x+y, xy \in \overline{A}$  □

In particular,  $\overline{\mathbb{Z}}$  is a ring.

**Corollario 11.** Take  $A \subseteq B \subseteq C$  a tower of inclusion of domains. Assume  $C$  is integral over  $b$  and  $B$  is integral over  $A$ . Then  $C$  is integral over  $A$ .

*Dimostrazione.* Take  $x \in C$ . Let  $f \in B[X]$  monic with  $f(x) = 0$  let  $b_1, \dots, b_n$  be the coefficients of  $f$  and set  $B' = A[b_1, \dots, b_n] \subseteq B$ . Since  $b_1, \dots, b_n$  are integral over  $A$ ,  $B'$  is a finitely generated  $A$ -module. Since  $x$  is integral over  $B'$ ,  $B'[x]$  is a finitely generated  $B'$ -module. Hence,  $B'[x]$  is finitely generated as  $A$ -module, so  $x$  is integral over  $A$ .  $\square$

In particular, if  $\overline{A} = \{x \in B : x \text{ integral over } A\}$ , then  $\{x \in C : x \text{ integral over } A\} = \{x \in C : x \text{ integral over } \overline{A}\}$

### Definizione 12

Let  $A \subseteq B \subseteq C$  domains.

1. The **integral closure** of  $A$  in  $B$  is the ring

$$\overline{A} = \{x \in B : x \text{ integral over } A\}$$

2. If  $B = \mathcal{Q}(A)$ , then  $\overline{A}$  is called the **integral closure of  $A$** .
3. If  $\overline{A} = A$  we say that  $A$  is **integrally closed** or **normal**
4. If  $k$  is a finite extension of  $\mathbb{Q}$  (i.e. a number field) then the integral closure of  $\mathbb{Z}$  in  $k$  is called the **ring of integers** of  $k$  and is denoted  $\mathcal{O}_k := i(K) \cap \overline{\mathbb{Z}}$  with  $i : K \hookrightarrow \mathbb{C}$  injective.

**Esempio 13.** UFDs are integrally closed. In particular,  $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$

**Esempio 14.** If  $k = \mathbb{Q}(\sqrt{d})$  with  $d \neq 0, 1$  squarefree, then  $\mathbb{Z}[\sqrt{d}] \subseteq \mathcal{O}_k$ . In general (depending on the value of  $d$ ),  $\mathbb{Z}[\sqrt{d}]$  is not necessarily equal to  $\mathcal{O}_k$ . If  $d \equiv 1 \pmod{4}$  then  $\frac{1+\sqrt{d}}{2} \in \mathcal{O}_k \setminus \mathbb{Z}[\sqrt{d}]$

**The Reference Framework (RF)** is having

- $A$  a domain with  $\mathcal{Q}(A) = K$
- $L/K$  a finite separable extension
- $B \subseteq L$  the integral closure of  $A$  in  $L$

In particular, we are interested in the cases with  $A = \mathbb{Z}$  and  $K = \mathbb{Q}$

## 1.2 How to compute rings of integers

During this section we work with RF.

**Proposizione 15.** Take  $x \in L$ . Then  $x \in B$  if and only if the minimal polynomial  $\mu$  of  $x$  has coefficient in  $A$ .

*Dimostrazione.*

$\Rightarrow$  Assume  $x \in B$ . Take  $f \in A[X]$  monic with  $b(x) = 0$ . Then  $\mu|f$  in  $K[X]$ . So the roots  $z_1, \dots, z_n$  of  $\mu$  are roots of  $f$ . Hence,  $z_1, \dots, z_n$  are integral over  $A$ . But the coefficients of  $\mu$  are in  $A[z_1, \dots, z_n]$  hence they are integral over  $A$ . But they are in  $K$  and  $A$  is integrally closed, hence they are in  $A$  and  $\mu \in A[X]$

$\Leftarrow$  obvious

□

**Corollario 16.** Take  $A = \mathbb{Z}$ ,  $K = \mathbb{Q}$  and  $L = \mathbb{Q}(\sqrt{d})$  with  $d \neq 0, 1$  squarefree. Then

$$\mathcal{O}_L = \begin{cases} \mathbb{Z}[\sqrt{d}] & d \equiv 2, 3 \pmod{4} \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] & d \equiv 1 \pmod{4} \end{cases}$$

*Dimostrazione.* Take  $a+b\sqrt{d} \in L$ , with  $a, b \in \mathbb{Q}$  and  $b \neq 0$ . The minimal polynomial is

$$\mu = (X - a)^2 - b^2 d = X^2 - 2aX + a^2 - db^2$$

Hence  $x \in \mathcal{O}_L \iff \mu \in \mathbb{Z}[X] \iff 2a \in \mathbb{Z}$  and  $a^2 - db^2 \in \mathbb{Z}$ . If  $x \in \mathcal{O}_L$ , then

$$(2a)^2 - 4(a^2 - db^2) = 4db^2 = d(2b)^2$$

Since  $d$  is squarefree,  $2b \in \mathbb{Z}$ . Write  $a = \frac{m}{z}$ ,  $b = \frac{n}{z}$  with  $m, n \in \mathbb{Z}$ . We conclude in the two cases with the thesis. □

We introduce some algebraic tools.  $L/K$  is finite separable. By the primitive element theorem we know  $L = K(\alpha)$ , with  $\alpha \in L$  and  $L = K(\alpha) \cong K[X]/(\mu)$  where  $\mu$  is the minimal polynomial of  $\alpha$ .

Take  $L^s$  a separable closure of  $L$ .  $\text{Hom}_K(L, L^s) = \{\sigma : L \hookrightarrow L^s : \sigma|_K = \text{id}_K\} \cong \{\text{roots of } \mu \text{ in } L^s\}$ . In particular,  $|\text{Hom}_K(L, L^s)| = [L : K]$ . Coming back to the question of finding out if some elements of  $L$  are integral or not over  $A$ , we know how to compute polynomials that vanish at some  $x \in L$ :

$$\begin{aligned} m_x : L &\longrightarrow L \\ y &\longmapsto m_x(y) = xy \end{aligned}$$

and  $\chi_{m_x}(x) = 0$

### Definizione 17

$\chi_{m_x} =: \chi_x$  is called the characteristic polynomial of  $x$ .

$$\chi_x = X^n - a_{n-1}X^{n-1} + \cdots + (-1)^n a_0$$

and  $a_{n-1} = \text{Tr}_{L/K}(x)$  the trace of  $x$ ,  $a_0 = N_{L/K}(x)$  the norm of  $x$ .

**Proposizione 18.** We have:  $\mu_x$  minimal polynomial of  $x$ . Then

$$\chi_x = (\mu_x)^{[LK(x)]} = \prod_{\sigma \in \text{Hom}_K(L, L^s)} (X - \sigma(x))$$

and

$$\text{Tr}_{L/K}(x) = \sum_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x) \quad ; \quad N_{L/K}(x) = \prod_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x)$$

*Dimostrazione.*

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_{n-1} \\ 1 & 0 & \cdots & 0 & -a_{n-2} \\ 0 & 1 & \cdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \cdots & 1 & -a_0 \end{pmatrix}$$

where  $\mu = X^n + a_{n-1}X^{n-1} + \dots + a_0$ . Since the characteristic poly of tthis matrix is  $\mu$ , we get

$$\chi_x = \mu_x^v = \mu_x^{[L:K(x)]}$$

We know:

$$\begin{aligned}\mu_x &= \prod_{\tau \in \text{Hom}_K(K(x), L^s)} (X - \tau(x)) \\ \chi_x &= \prod_{\tau \in \text{Hom}_K(K(x), L^s)} (X - \tau(x))^{[L:K(x)]}\end{aligned}$$

We have to check that, for each  $\tau \in \text{Hom}_K(K(x), L^s)$ , there exists exactly  $[L : K(x)]$  elements  $\sigma \in \text{Hom}_K(L, L^s)$  with  $\sigma|_{K(x)} = \tau$ .

Then, by PET,  $L = K(x, y)$  with some  $y \in L$ . Let  $\mu_y$  be the minimal polynomial of  $y$  over  $K(x)$ . Fixing  $\tau \in \text{Hom}_K(K(x), L^s)$ ,

$$\text{Hom}_{K(x)}(L, L^s) = \{\sigma : L \hookrightarrow L^s : \sigma|_{K(x)} = \tau\} \text{ has } [L : K(x)] = \deg \mu_y \text{ elements}$$

□

**Corollario 19.** Take  $x \in L$ . Then  $x \in B$  if and only if  $\chi_x \in A[X]$ . In particular, if  $x \in B$ , then  $\text{Tr}_{L/K}(x) \in A$  and  $N_{L/K}(x) \in A$

*Dimostrazione.*

□

Fix  $\mathbf{x} = (x_1, \dots, x_n)$  a  $K$ -basis of  $L$ . Given  $x = \sum_{i=1}^n d_i x_i$ ,  $d_i \in K$ , I want to express in terms of the  $d_i$ 's whether  $x \in B$  or  $x \notin B$ .

**Lemma 20.**  $\mathcal{Q}(B) = L$ . Better, for each  $x \in L$ , there exists  $y \in A \setminus \{0\}$  such that  $yx \in B$ .

*Dimostrazione.* Take  $\mu_x \in K[X]$  the minimal polynomial of  $x$ . Write  $a_i = b_i/c_i$ , with  $b_i, c_i \in A$ . Set  $c := (c_0, \dots, c_{m-1})$  and  $z := cx$ . Then

$$\left(\frac{z}{c}\right)^m + a_{m-1}\left(\frac{z}{c}\right)^{m-1} + \dots + a_0 = 0$$

by multiplying everything times  $c^m$  we find  $z \in B$ , hence  $x = \frac{z}{c}$  with  $z \in B$  and  $c \in A$ . □

Up to multiplying  $\mathbf{x}$  by some well-chosen  $a \in A$ , we may assume  $x_1, \dots, x_n \in B$ . We come back to

$$x = \sum_{i=1}^n \lambda_i x_i, \quad \lambda_i \in K$$

If  $x \in B$ , then  $\forall i, x x_i \in B_n$  and  $\text{Tr}_{L/K}(x x_i) = \sum_{j=1}^n \lambda_j \text{Tr}_{L/K}(x_i x_j) \in A$ .

If we set  $M_{\mathbf{x}} = (\text{Tr}_{L/K}(x_i x_j)) \in \mathcal{M}_n(A)$ , then  $M_{\mathbf{x}} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} \in A^n$  and hence its

determinant is in  $A^n$ .

**Proposizione 21.** If  $d(\mathbf{x}) = \det(M_{\mathbf{x}}) = \det(\text{Tr}_{L/K}(x_i x_j))$  and  $d(\mathbf{x})B \subseteq \sum_{i=1}^n A x_i$

#### Definizione 22: discriminant

$d\mathbf{x}$  is called the **discriminant** of  $\mathbf{x}$

## Computation of the discriminant

$$\begin{aligned}
d(\mathbf{x}) &= \det(\mathrm{Tr}_{L/K}(x_i x_j)) = \\
&= \det\left(\sum_{\sigma \in \mathrm{Hom}_K(L, L^s)} \sigma(x_i) \sigma(x_j)\right) = \\
&= \left(\det(\sigma(x_j))_{\sigma, j}\right)^2
\end{aligned}$$

### 1.2.1 Theoretical algorithm to compute rings of integer

Take  $A = \mathbb{Z}$ ,  $K = \mathbb{Q}$ ,  $L$  a number field and  $B = \mathcal{O}_L$ , then

- 1) Find a  $\mathbb{Q}$ -basis  $\mathbf{x} = (x_1, \dots, x_n)$  of  $L$  contained in  $B$  e.g.  $\mathbf{x} = (1, x, x^2, \dots, x^{n-1})$  with  $x \in B$  a primitive element for  $L/\mathbb{Q}$
- 2) Compute  $d(\mathbf{x})$
- 3) Say that

$$\bigoplus_i \mathbb{Z}x_i \subseteq B \subseteq \frac{1}{d(\mathbf{x})} \bigoplus_i \mathbb{Z}x_i$$

- 4) Compute representation  $y_1, \dots, y_n$  of the quotient

$$\left(\frac{1}{d(\mathbf{x})} \bigoplus_i \mathbb{Z}x_i\right) / \left(\bigoplus_i \mathbb{Z}x_i\right)$$

**Proposizione 23.** *In the RF. Assume  $A$  is a PID. Then*

1. *There exists a  $K$ -basis  $\mathbf{x} = (x_1, \dots, x_n)$  of  $L$  contained in  $B$  such that*

$$B = \bigoplus_i Ax_i$$

*such a basis  $\mathbf{x}$  is called an **integral basis** of  $B$*

2. *Assume further that  $A = \mathbb{Z}$ . Then  $d(\mathbf{x})$  is independant of the choice of an integral basis  $\mathbf{x}$  and if  $\mathbf{y}$  is any  $K$ -basis of  $L$  contained in  $B$ , then*

$$d(\mathbf{y}) = [B : \Gamma_{\mathbf{y}}]^2 \cdot d(\mathbf{x})$$

*Osservazione.* If  $A = \mathbb{Z}$  and  $K = \mathbb{Q}$ . Say you have a  $\mathbb{Q}$ -basis  $\mathbf{y}$  of  $L$  contained in  $\mathcal{O}_L$  such that  $d(\mathbf{y})$  is squarefree. Then  $\mathcal{O}_L = \bigoplus \mathbb{Z}y_i$

*Dimostrazione.* 1) Take  $\mathbf{y}$  a  $K$ -basis of  $L$  contained in  $B$ . We know

$$\bigoplus_i Ay_i \subseteq B \subseteq \frac{1}{d(\mathbf{y})} \bigoplus_i Ay_i$$

By classication of finitely generated modules over PIDs,  $B$  is a free  $A$ -module of rank  $[L : K]$ . Hence, there exists  $\mathbf{x} = (x_1, \dots, x_n)$  such that  $B = \bigoplus_i Ax_i$ . Necessarily,  $\mathbf{x}$  is also a  $K$ -basis of  $L$

- 2) Let  $\mathbf{x}$  be an integral basis of  $B = \mathcal{O}_L$ . Let  $\mathbf{y}$  be any  $\mathbb{Q}$ -basis of  $L$  contained in  $\mathcal{O}_L$ . We know that  $d(\mathbf{x})$  (resp.  $d\mathbf{y}$ ) is the determinant of the trace form  $\mathrm{Tr}_{L/K}$  on the basis  $\mathbf{x}$  (resp.  $\mathbf{y}$ ). If  $P$  is the transition matrix from  $\mathbf{x}$  to  $\mathbf{y}$ , then  $d(\mathbf{y}) = d(\mathbf{x})(\det P)^2$ . By Smith's nor form theorem (théorème de la base adaptée),  $|\det P| = [\bigoplus_i \mathbb{Z}x_i : \bigoplus_i \mathbb{Z}y_i] = [\mathcal{O}_L : \Gamma_{\mathbf{y}}]$ . We conclude with the equivalence in the thesis. Moreover, if  $\mathbf{y}$  was an integral basis, then  $\Gamma_{\mathbf{y}} = \mathcal{O}_L$  and thus  $d(\mathbf{y}) = d(\mathbf{x})$

□

#### Definizione 24

Let  $L$  be a number field. The discriminant of an integral basis of  $L$  is called **discriminant** of  $L$  and it's noted  $\Delta_L$

### 1.3 Algebraic properties

Rings of integers do not satisfy many properties. They are not PID generally for instance.

#### Definizione 25: Dedekind domains

A domain  $A$  is said to be a **Dedekind domain** if it is Noetherian, integrally closed and every non-zero prime ideal in  $A$  is maximal. Equivalently, it's a Noetherian domain with Krull dimension at most 1.

**Esempio 26.** PIDs are Dedekind domains. UFDs are not all Dedekind domains. For instance, in  $A = \mathbb{C}[x, y]$   $(x)$  is not maximal.

#### Teorema 27

Working in the Framework RF (even though separability of  $L/K$  is not needed). Assume  $A$  is a Dedekind domain. Then  $B$  is also a Dedekind domain.

*Osservazione.* In particular, rings of integers in number fields are Dedekind domains.

*Dimostrazione.* (We use the assumption of separability of  $L/K$ ). We already know that  $B$  is integrally closed. We know that, if  $\mathbf{x} = (x_1, \dots, x_n)$  is a  $K$ -basis of  $L$  contained in  $B$  and  $B \subseteq \frac{1}{d(\mathbf{x})} \bigoplus A x_i$  which is a finitely generated  $A$ -module, hence noetherian. It follows that  $B$  is a noetherian ring.

Take a non-zero prime ideal  $\mathfrak{p}$  in  $B$ . Set  $\mathfrak{q} = \mathfrak{p}^c = \mathfrak{p} \cap A$ . Then  $\mathfrak{q}$  is a prime ideal in  $A$ . Moreover, if one takes  $x \in \mathfrak{p} \setminus \{0\}$ , then  $x$  satisfies some equation

$$x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0 \quad a_i \in A, a_0 \neq 0$$

Hence

$$a_0 = -x^n - a_{n-1}x^{n-1} - \dots - a_1x \in \mathfrak{p} \cap A = \mathfrak{q}$$

thus  $\mathfrak{q}$  is non-zero, hence maximal.  $B/\mathfrak{p}$  is a domain which is also a finite-dimensional  $(A/\mathfrak{q})$ -vector space ( $B$  is finitely generated  $A$ -module). Hence,  $B/\mathfrak{p}$  is a field and  $\mathfrak{p}$  is maximal.  $\square$

#### 1.3.1 Fractional ideals

#### Definizione 28

Let  $A$  be a domain with  $\mathcal{Q}(A) = K$ . A fractional ideal  $I$  in  $K$  is an  $A$ -submodule of  $K$  such that there exists  $a \in A \setminus \{0\}$  satisfying  $aI \subseteq A$

**Esempio 29.** If  $A$  is a PID, fractional ideals are the  $xA$  with  $x \in K \setminus \{0\}$ . In particular if  $A$  is not a field,  $K$  is not a fractional ideal.

#### Teorema 30

Let  $A$  be a Dedekind domain. All non-zero prime ideals in  $A$  are invertible.

Take  $\mathfrak{p}$  a non-zero prime ideal in  $A$ . If  $J$  is a fractional ideal such that  $\mathfrak{p}J \subseteq A$ , then we define

$$J \subseteq \{x \in K : x\mathfrak{p} \subseteq A\} =: \mathfrak{p}^{-1}$$

Note that if  $y \in \mathfrak{p} \setminus \{0\}$ , then  $y\mathfrak{p}^{-1} \subseteq A$ , hence  $\mathfrak{p}^{-1}$  is a fractional ideal.

The theorem will then follow from the following

**Proposizione 31.** *Let  $A$  be a Dedekind domain,  $\mathfrak{p}$  a non-zero prime ideal in  $A$  and  $I$  any non-zero ideal in  $A$  and  $I$  any non-zero ideal in  $A$ . Then  $I \not\subseteq \mathfrak{p}^{-1}I$ . In particular,  $\mathfrak{p}^{-1}\mathfrak{p} = A$*

*Osservazione.* In particular,  $\mathfrak{p} \subsetneq \mathfrak{p}^{-1}\mathfrak{p} \subseteq A$  since  $\mathfrak{p}$  is maximal,  $\mathfrak{p}^{-1}\mathfrak{p} = A$

*Dimostrazione.* *Step 1:* We prove that every nonzero ideal  $I \subseteq A$  contains a product of nonzero prime ideals. Assume by contradiction this is wrong. Consider  $\mathcal{E}$  a set of nonzero ideals  $I \subseteq A$  that do not contain a product of nonzero prime ideals. By assumption  $\mathcal{E} \neq \emptyset$  and using Notherianity, it has a maximal element  $I_0$ .  $I_0$  is not prime, so we can find  $a, b \in A \setminus I_0$  such that  $ab \in I_0$ . Then  $I_0 + (a)$  and  $I_0 + (b)$  both contain product of nonzero prime ideals. Hence,  $(I_0 + (a))(I_0 + (b)) \subseteq I_0$  contains a product of nonzero prime ideals, which leads to a contradiction.

*Step 2:* We prove that  $A \subsetneq \mathfrak{p}^{-1}$ .  $\supseteq$  is obvious. Take now  $a \in \mathfrak{p} \setminus \{0\}$ ; by *Step 1* we can write  $\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq (a) \subseteq \mathfrak{p}$  and we suppose  $r$  as small as possible, with all  $\mathfrak{p}_i$  prime ideals. There exists  $i_0$  such that  $\mathfrak{p}_{i_0} \subseteq \mathfrak{p}$  and we may assume  $i_0 = r$ . Since  $\mathfrak{p}_r$  is maximal,  $\mathfrak{p}_r = \mathfrak{p}$ . Moreover, since  $r$  is as small as possible, we can find  $b \in \mathfrak{p}_1 \dots \mathfrak{p}_{r-1} \setminus (a)$ . Then  $a^{-1}b \notin A$  and  $\forall x \in \mathfrak{p}$ ,  $a^{-1}bx \in a^{-1}\mathfrak{p}_1 \dots \mathfrak{p}_{r-1}\mathfrak{p} = a^{-1}\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq a^{-1}(a) = A$ . Hence,  $a^{-1}b \in \mathfrak{p}^{-1}$ .

*Step 3:* We prove the proposition  $\mathfrak{p}^{-1}I \not\supseteq I$  by contradiction. Assume  $\mathfrak{p}^{-1}I = I$  and take  $x \in \mathfrak{p}^{-1}$ . Then  $x \in \mathfrak{p}^{-1}$ . Then  $xI \subseteq I$  and by Notherianity of  $A$ ,  $I$  is finitely generated. By a previous proposition, this allows to find a monic  $\chi \in A[X]$  such that  $\chi(x) = 0$ . But  $A$  is integrally closed, hence  $x \in A$ . Hence  $\mathfrak{p}^{-1} = A$  which contradicts *Step 2*.  $\square$

### 1.3.2 Factorization of ideals

#### Teorema 32

Let  $A$  be a Dedekind domain. Then every nonzero ideal of  $A$  can be written as a product of nonzero prime ideals, in a unique way up to the order.

*Dimostrazione. Existence:* Let  $\mathcal{E}$  be the family of ideals in  $A$  other than  $(0)$  and  $(1)$  that are not product of prime ideals. By contradiction, assume  $\mathcal{E} \neq \emptyset$ . By Notherianity, we can find a maximal element  $I_0 \in \mathcal{E}$ . Since  $I_0 \neq A$  we can find a maximal ideal  $\mathfrak{p}$  of  $A$  containing  $I_0$ . Set  $J = \mathfrak{p}^{-1}I_0$ .

Then

$$i_0 \subsetneq \mathfrak{p}^{-1}I_0 \subseteq \mathfrak{p}^{-1}\mathfrak{p} = A$$

So  $\mathfrak{p}^{-1}I_0$  is an ideal of  $A$  not in  $\mathcal{E}$ . So  $\mathfrak{p}^{-1}I_0 = \mathfrak{p}_1 \dots \mathfrak{p}_r$  for some prime ideals  $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ . We conclude  $I_0 = \mathfrak{p}\mathfrak{p}^{-1}I_0 = \mathfrak{p}\mathfrak{p}_1 \dots \mathfrak{p}_r$ .  $\square$

## 1.4 Ramification

### Definizione 33

In the RF with  $A$  Dedekind, take  $\mathfrak{p}$  a prime ideal in  $A$ . Write the factorization of  $\mathfrak{p}B = p_1^{e_1} \dots p_g^{e_g}$ .

1.

**Proposizione 34.** *Almost all prime ideals of  $A$  are unramified in  $L$ .*

*Dimostrazione.* Let  $\alpha \in B$  a primitive element for  $L/K$  and take  $\mu$  its minimal polynomial. Set  $\square$

**Teorema 35**

Consider the *discriminant ideal*  $\mathfrak{d}_{B/A}$  of  $A$  generated by the discriminants of all  $K$ -bases of  $L$  contained in  $B$ .

A prime ideal  $\mathfrak{p}$  of  $A$  is ramified in  $L \iff \mathfrak{p} | \mathfrak{d}_{B/A}$

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